

# Florentin Coeurdoux

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## RESEARCH INTERESTS

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Generative models — flows, stochastic interpolants, and energy-based models — Bayesian inference, and optimal transport. Building principled methods that connect rigorous theory with large-scale real-world deployment.

## WORK EXPERIENCE

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### Capital Fund Management (CFM)

Paris, France

*VP, Quantitative Research*

*Jan 2024 – Present*

- Lead a team of quantitative researchers building and deploying ML models for live alpha generation on futures markets, covering commodities, equity indices, FX, bonds, short-term interest rates, and volatility products.
- Own the full pipeline from signal research to production deployment at one of the world's leading systematic hedge funds.

*Research Scientist, CFM AI Lab*

*Jan 2024 – Present*

- Conduct research on generative models and statistical inference — spanning flows, stochastic interpolants, and energy-based models — in close collaboration with E. Vanden-Eijnden (Courant/NYU) and S. Mallat (ENS/Flatiron Institute).
- Published 6 papers in 2025–2026, including at NeurIPS 2025 and ICML 2026; co-investigated applications of random matrix theory to ML with J.-P. Bouchaud (CFM).
- Supervise PhD students and research interns.

### University of Oxford, Department of Statistics

Oxford, UK

*Visiting Research Scholar*

*Oct 2022 – Mar 2023*

- Conducted research on Monte Carlo methods and deep generative models under the supervision of Prof. Arnaud Doucet.

### AssessFirst

Paris, France

*Head of Data Science*

*Dec 2021 – Nov 2022*

- Led all data science initiatives and set the AI/ML roadmap for the company.
- Developed a variational inference algorithm served to 100k users, accelerating inference by 30%.
- Designed a statistical pattern detection algorithm for hiring, securing a 1.1M€ contract.

### Credit Mutuel Arkéa

Rennes, France

*Quantitative Developer*

*Apr 2019 – Nov 2020*

- Developed a multivariate time series forecasting model for overdraft risk, deployed in production for 1.2 million users.
- Migrated option pricing codebase from VBA to C++, reducing execution time by 25% via importance sampling.

## EDUCATION

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### Institut National Polytechnique (INP) – ANITI

Toulouse, France

*Ph.D. in Applied Mathematics and Statistics*

*Nov 2020 – Dec 2023*

- Supervised by Prof. Nicolas Dobigeon (IRIT/INP) and Prof. Pierre Chainais (Centrale Lille). Thesis on Monte Carlo sampling and deep generative models for Bayesian inference.

## PUBLICATIONS

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- 📄 E. Moreau, **F. Coeurdoux**, G. Ferré, E. Vanden-Eijnden, “Probing the Geometry of Diffusion Models with the String Method”, *ICML*, 2026. [arXiv]
- 📄 H. Negrel, **F. Coeurdoux**, M. Albergo, E. Vanden-Eijnden, “Multitask Learning with Stochastic Interpolants”, *NeurIPS*, 2025. [arXiv]
- 📄 **F. Coeurdoux**, E. Lempereur, N. Cuvelles-Magar, S. Mallat, E. Vanden-Eijnden, “Generative Modeling via Kernelized Stochastic Interpolants”, *ICML Workshop*, 2026. [arXiv]
- 📄 E. Lempereur, N. Cuvelles-Magar, **F. Coeurdoux**, S. Mallat, E. Vanden-Eijnden, “MGD: Moment Guided Diffusion for Maximum Entropy Generation”, *MFML Workshop*, 2026. [arXiv]
- 📄 M. Chalvidal, **F. Coeurdoux**, E. Vanden-Eijnden, “Covariance Shrinkage via Stochastic Interpolation”, *Preprint*, 2026. [arXiv]
- 📄 **F. Coeurdoux**, G. Ferré, J.-P. Bouchaud, “Random Matrix Theory of Early-Stopped Gradient Flow”, *Preprint*, 2026. [arXiv]
- 📄 **F. Coeurdoux**, N. Dobigeon, P. Chainais, “Plug-and-Play Split Gibbs Sampler: Embedding Deep Generative Priors in Bayesian Inference”, *IEEE Transactions on Image Processing*, 2024. [arXiv]
- 📄 **F. Coeurdoux**, N. Dobigeon, P. Chainais, “Normalizing Flow Sampling with Langevin Dynamics in the Latent Space”, *Machine Learning*, 2023. [arXiv]
- 📄 **F. Coeurdoux**, N. Dobigeon, P. Chainais, “Learning Optimal Transport Between Two Empirical Distributions with Normalizing Flows”, *ECML-PKDD*, 2022. [arXiv]
- 📄 **F. Coeurdoux**, N. Dobigeon, P. Chainais, “Sliced-Wasserstein Normalizing Flows: Beyond Maximum Likelihood Training”, *ESANN*, 2022. [arXiv]

## SERVICE

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**Reviewer:** NeurIPS, ICML, ICLR, AISTATS.

## SKILLS

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**Programming:** Python, C++.

**ML/DL:** PyTorch, JAX, CUDA.

**Infrastructure:** Distributed Training, Docker, Git, Linux, AWS.

**Languages:** French (Native), English (Fluent).

## REFERENCES

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Available upon request.